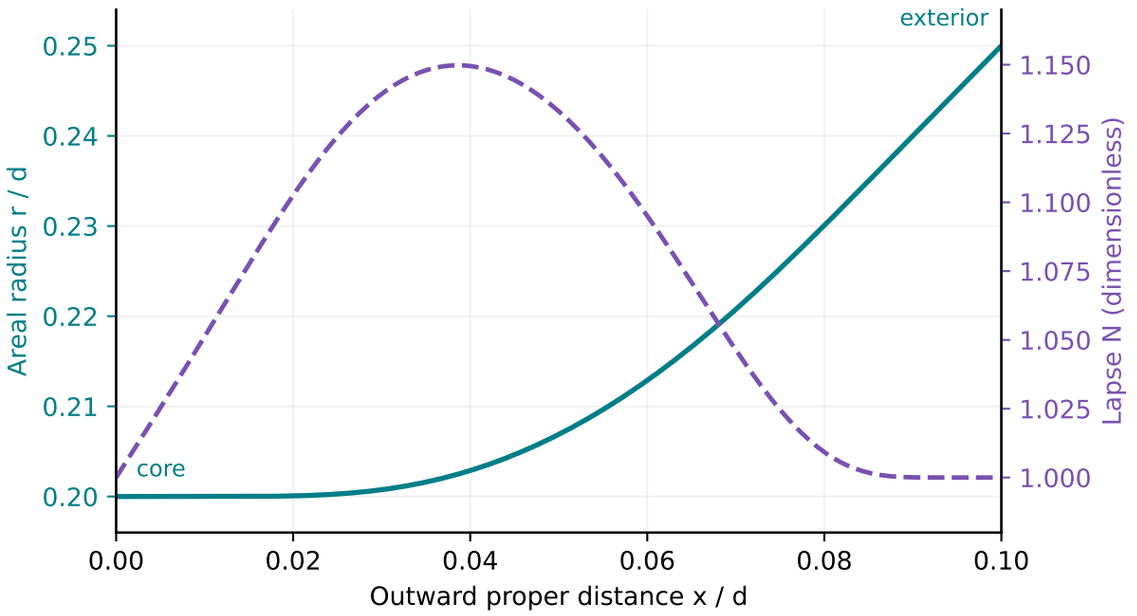
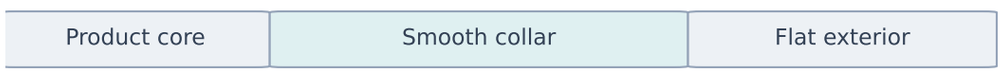
A smooth collar still needs its supporting source

One prescribed collar: radius 0.20d → 0.25d over proper width 0.10d · Smooth positive lapse · q = 1

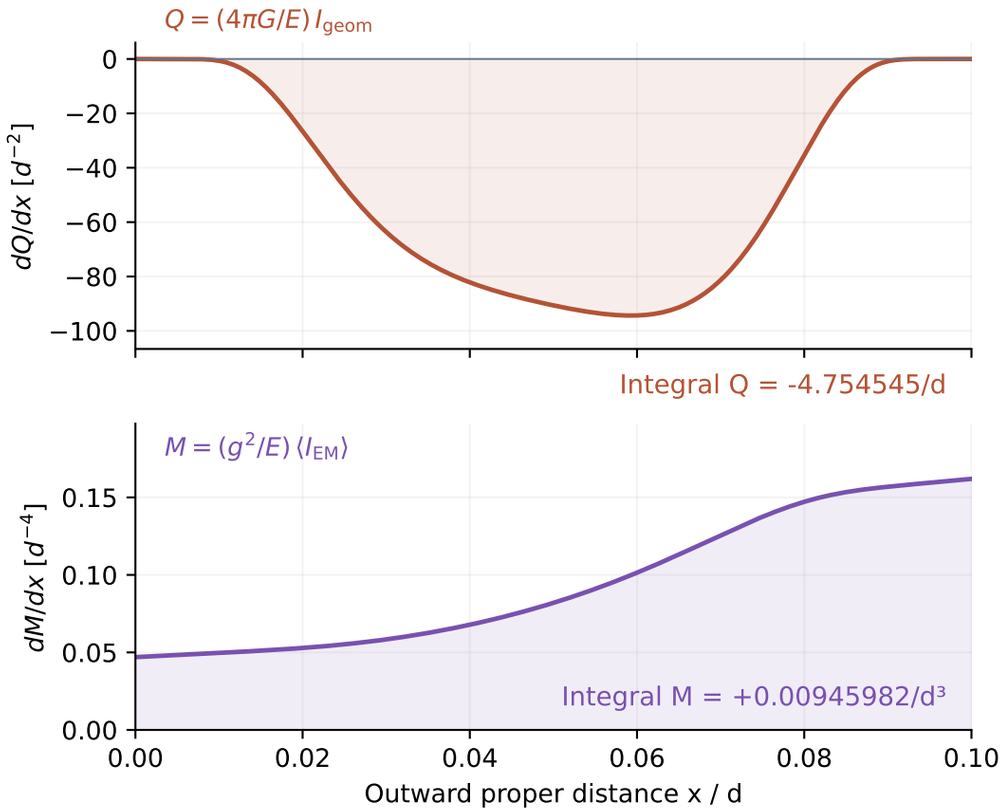
A. Actual prescribed radius and clock-rate profiles





Region labels only · one-collar optical length = 0.093438d

B. Required geometry and known magnetic term



Remaining source: $Q_{\mathrm{other}} = Q_{\mathrm{geom}} - (4\pi G/g^2)M$ **· G/g² stays symbolic; the plotted normalizations are not added.**

I is an affine radial-null stress integral; E is the conserved null-ray energy. The magnetic contribution is nonnegative.

L = 9 matching; modes through 15 tested. Higher tails, nonspherical Einstein balance, quantum support and stability remain unresolved.

Next: gravitational exterior matching. If $b = r_e$, the flat near-mouth approximation is uncontrolled ($r_e^2/r_{\mathrm{out}}^2 = 0.64$).

13 Sep 2027 → 12 Sep 2027 remains unproved; no physical medium or construction method is established.